VIS4AI / AI4VIS LAB · TOPIC SHORTLIST

Two project topic ideas: AI detectors under the microscope

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> **The detector is the object we study, not something we try to perfect.** In both topics we use visualization to understand *how* a detector behaves (Phase I, black box) and *why* (Phase II, looking inside). In parallel we run the AI4Vis experiment: one pair works with ChatGPT/Claude and one pair works without, and the pairs swap after Phase I.

Why these two

- Both run on **Intel Macs**. A free Kaggle GPU is only needed for optional fine-tuning.
- Public datasets with **group metadata**, which helps find systematic behavior rather than cherry-picked examples.
- A clear path from Phase I to Phase II: a behavior we find becomes a hypothesis we test inside the model.

Group set-up (both topics)

- **Phase I:** Pair A without GenAI, Pair B with GenAI.
- **Phase II:** the pairs swap. Each pair uses at least 2 explanation techniques.
- The pair working with GenAI keeps an **AI interaction log** (prompts, what was accepted, modified or rejected, failure patterns).

TOPIC 1 · TEXT

“Would an AI detector accuse you of cheating?”

Bias of AI-text detectors against English-language learners

THE IDEA

Schools and universities use AI-text detectors to catch essays written by ChatGPT. Liang et al. (2023) found that popular GPT detectors **consistently misclassify writing by non-native English speakers as AI-generated**, while writing by native speakers is identified correctly. Detectors tend to treat simple, predictable writing as “AI-like”, and many second-language writers write that way. We investigate *who* gets falsely flagged, *what* those essays have in common, and *what inside the detector* causes it.

MODELS

MODEL	ROLE	LINK
chatgpt-detector-roberta MAIN	Downloaded and used as-is. RoBERTa-base fine-tuned to detect ChatGPT answers (trained on the HC3 dataset).	huggingface.co/Hello-SimpleAI/chatgpt-detector-roberta
roberta-base-openai-detector OPTIONAL	A second detector for comparison. Older: trained to detect GPT-2 output.	huggingface.co/openai-community/roberta-base-openai-detector
distilbert-base-uncased OPTIONAL	Base model for fine-tuning **our own** detector. Gives known training data and training checkpoints (~1 h of free Kaggle GPU).	huggingface.co/distilbert/distilbert-base-uncased

DATASETS

DATASET	WHAT IT CONTAINS	LINK
PERSUADE 2.0 corpus MAIN	25,000+ argumentative essays by US students (grades 6–12, 15 prompts) with demographic information per writer, including English-learner status. Licence CC BY-NC-SA 4.0.	github.com/scrosseye/persuade_corpus_2.0
DAIGT V2 Train Dataset MAIN	Community dataset of human essays plus AI-generated essays from many different LLMs on the same prompts.	kaggle.com/datasets/thedrcat/daigt-v2-train-dataset
LLM – Detect AI Generated Text	Kaggle competition on the same essay prompts: context, discussions and additional data.	kaggle.com/competitions/llm-detect-ai-generated-text
Feedback Prize – English Language Learning	Backup or extra data: essays by English learners scored on vocabulary, grammar, syntax and more.	kaggle.com/competitions/feedback-prize-english-language-learning
HC3	The training data of the main detector. We explore it as a source of hypotheses (what "human" and "AI" text looked like in training).	huggingface.co/datasets/Hello-SimpleAI/HC3

Background paper: Liang et al., "GPT detectors are biased against non-native English writers" (2023), arxiv.org/abs/2304.02819

PHASE I: HOW DOES IT BEHAVE?

- **Who** gets falsely flagged? English learners vs. native speakers, grade level, essay prompt.
- **What** do falsely flagged human essays share? Vocabulary richness, sentence length, essay score.
- **What lets AI essays slip through?** Which LLM wrote them, typos, paraphrasing.

Visualizations: AI-score distributions split by group · score vs. text properties · 2D map of essays (UMAP of independent text embeddings) coloured by false positives, used to find clusters and sub-clusters · an essay reader for drill-down.

PHASE II: WHAT DID IT LEARN?

- **Word attributions** (Integrated Gradients, SHAP or LIME): which words push an essay toward "AI", compared in aggregate across falsely flagged and correct essays.
- **Representation maps per layer:** where do learner essays sit relative to AI essays inside the model?
- **Linear probes:** is "English proficiency" encoded, and is it entangled with "AI-ness"?
- **k-nearest neighbours** in the detector's representation space.
- **Checkpoints** (if we fine-tune our own): when does the bias appear?

Compute

Inference and attributions run on an Intel Mac CPU (roughly an hour for ~10k essays). Optional fine-tuning: ~1 h of free Kaggle GPU.

Risks and checks

The main detector was trained on Q&A answers (HC3), not essays, so it may behave oddly on essays. **Test on ~100 essays first.** If it is unusable, fine-tune our own detector on DAIGT data and use that as the main model.

TOPIC 2 · IMAGES

“What fools an AI-image detector?”

Robustness and shortcut learning in a real-vs-AI-generated image classifier

THE IDEA

Detectors are supposed to catch AI-generated images (deepfakes, fake news). They may rely on **invisible pixel-level traces** left by the generation process rather than on visible content. The CIFAKE authors’ own Grad-CAM analysis suggested that their model ignores the object itself and focuses on **small imperfections in the background**. If that is true, simple edits such as JPEG compression, resizing or blurring could erase the trace and let fakes pass. We test *what fools the detector*, and *what it really looks at*.

MODELS

MODEL	ROLE	LINK
Our own small CNN MAIN	**Trained by us** from scratch on CIFAKE (minutes to under an hour on a laptop). Known training data, and we save checkpoints during training. The CIFAKE paper describes a reference architecture.	`arxiv.org/abs/2303.14126` (reference architecture)
ResNet-18 (ImageNet-pretrained) OPTIONAL	Fine-tuned on CIFAKE as a second detector, to compare “pretrained” vs. “from scratch” (do both learn the same shortcut?).	`pytorch.org/vision/…/torchvision.models.resnet18`
UniversalFakeDetect (UnivFD) OPTIONAL	A published, CLIP-based detector for full-size images. Used for a comparison on a few larger fakes.	`github.com/WisconsinAIVision/UniversalFakeDetect`

DATASETS

DATASET	WHAT IT CONTAINS	LINK
CIFAKE MAIN	120,000 small (32×32) images: 60k real photos (from CIFAR-10) and 60k images generated with Stable Diffusion, in the same 10 classes (cats, cars, planes, ships…).	`kaggle.com/datasets/birdy654/cifake-real-and-ai-generated-synthetic-images`
Our own test variants SELF-MADE	CIFAKE test images after JPEG compression at several quality levels, resizing, blurring, added noise and screenshots. Optionally, fakes from a newer image generator.	generated with Python (Pillow / OpenCV)

Background paper: Bird & Lotfi, “CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images”, `arxiv.org/abs/2303.14126`

PHASE I: HOW DOES IT BEHAVE?

- **How fast does detection collapse** under JPEG, blur, resizing and noise? Shown as robustness curves per class.
- **Which classes are hardest?** Which *real* photos get flagged as fake, and what do they share (blur, brightness, texture)?
- **Training data:** how do real and fake images differ in simple statistics (sharpness, colour, noise)?

Visualizations: accuracy vs. edit strength · score distributions per class · 2D map of images (UMAP of independent image embeddings) coloured by errors, used to find clusters and sub-clusters · an error gallery for drill-down.

PHASE II: WHAT DID IT LEARN?

- **Grad-CAM saliency:** object vs. background. Aggregate the share of saliency that falls on the background, comparing correct and fooled images.
- **Frequency analysis:** Fourier spectra of real vs. fake images, and low-pass/high-pass filter tests. Does it rely on fine pixel detail?
- **Representation maps per layer:** where do real, fake and edited images separate?
- **k-nearest neighbours** to training images.
- **Checkpoints:** when during training does the shortcut appear?

Compute

Runs **fully on an Intel Mac CPU**, including training. No cloud needed.

Risks and checks

32×32 images are tiny, so they look pixelated on slides and limit how far results generalize. The detector may be very accurate on clean CIFAKE images, so the interesting behavior will be in the **edited variants**.

Side-by-side comparison

	TOPIC 1 · TEXT DETECTOR	TOPIC 2 · IMAGE DETECTOR
Main theme	Fairness: who gets falsely accused?	Robustness: what fools it, and what does it look at?
Main model	Downloaded (HuggingFace)	Trained by us
Runs on Intel Macs	Mostly (fine-tuning on free Kaggle GPU)	Fully, including training
Relatable for the audience	★★★ every student cares	★★☆
Visual appeal	Highlighted text, distributions	Images, heatmaps, frequency spectra
Main weakness	Text is less flashy on slides; domain mismatch of the detector	Tiny 32×32 images

Shared tools (free)

PURPOSE	TOOL	LINK
Free GPU when needed	Kaggle Notebooks (roughly 30 GPU hours per week per account; the quota can change)	`kaggle.com/code`
Gradient attributions	Captum (PyTorch): Integrated Gradients, saliency	`captum.ai`
Grad-CAM	pytorch-grad-cam	`github.com/jacobgil/pytorch-grad-cam`
SHAP	shap	`github.com/shap/shap`
LIME	lime	`github.com/marcotcr/lime`
2D maps	umap-learn	`umap-learn.readthedocs.io`
Interactive dashboards	Streamlit	`streamlit.io`

> **Intel Mac set-up note:** PyTorch stopped publishing Intel-Mac builds after version 2.2. Install `torch==2.2.2` together with `numpy<2` to avoid compatibility errors.

Deadlines

DATE	DELIVERABLE
21.10.2026	Register the group
28.10.2026	Register task, model and dataset (check the list of other groups' topics and register early)
11.11.2026	Midpoint (end of Phase I): code from both pairs, AI interaction log, progress video of at most 5 minutes
09.12.2026	Final (end of Phase II): all code, AI logs from both phases, 10-minute recorded presentation and slide deck

Next step: 30-minute feasibility check

- **Topic 1:** load the detector and run it on ~100 PERSUADE essays (learners and natives). Does it produce varied scores, with some false positives?
- **Topic 2:** train a small CNN on a CIFAKE subset for one epoch, then JPEG-compress the test images. Does accuracy drop?

All URLs were checked and resolved on 19 Oct 2026. Kaggle datasets and competitions need a free Kaggle account, and competition data requires accepting the competition rules. Dataset licences apply. PERSUADE 2.0 is CC BY-NC-SA 4.0 (non-commercial, which is fine for coursework).